# EP-0163

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## Ultra Thin Ice Tower Cooler for Raspberry Pi 4B

- Purchase URL [ https://52pi.com/products/52pi-ultra-thin-ice-tower-cooler-cooling-fan-for-raspberry-pi-4-model-b-cpu-fan ]


## Description

This is an ultra-thin ice tower cooler that only supports Raspberry Pi 4B. It includes ultra-thin fans and supports fan speed regulation. The speed can be controlled by programming the GPIO pins of the Raspberry Pi, and the heat sink is ultra-thin.

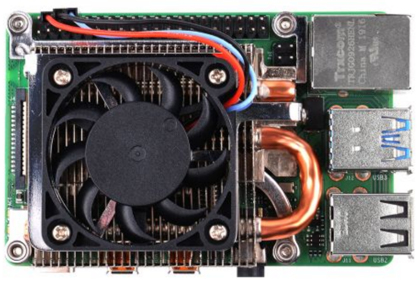
## Features

- Ultra Thin
- Support fan speed regultion
- Support Raspberry Pi 4B ONLY
- Good Heat dissipation effect
- 4010 Fan with PWM adjustable speed feature

## Gallery

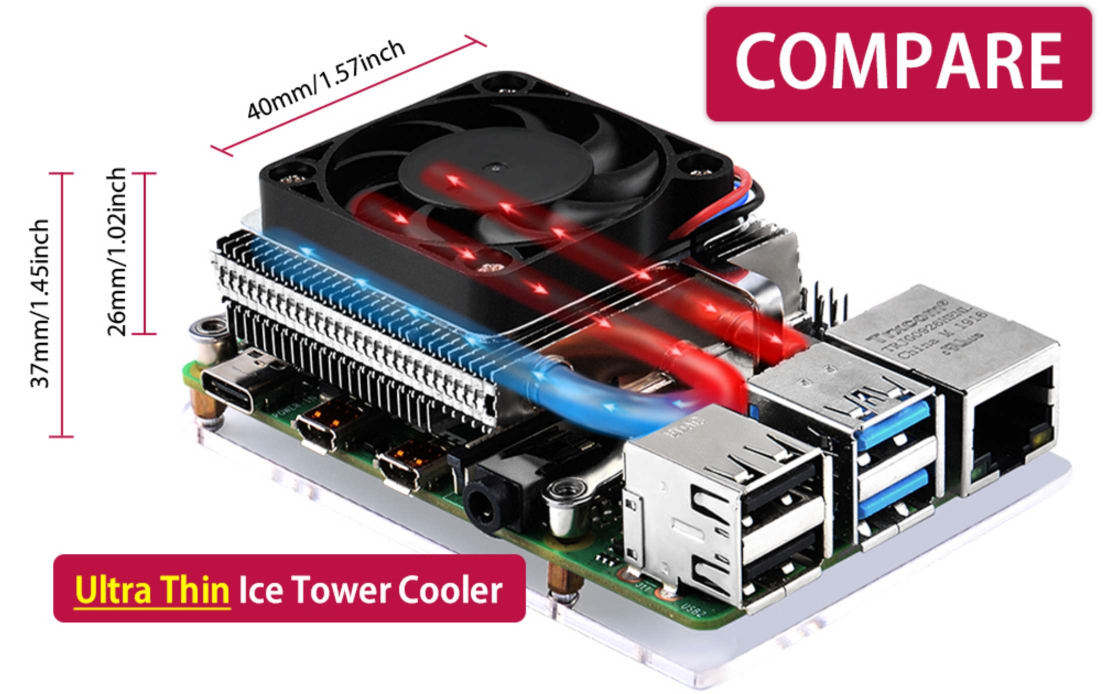
```
NOTE: Raspberry Pi 4B is not included in the package, need additional purchase.
```

- Product Outlook

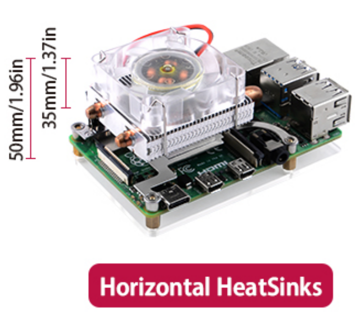



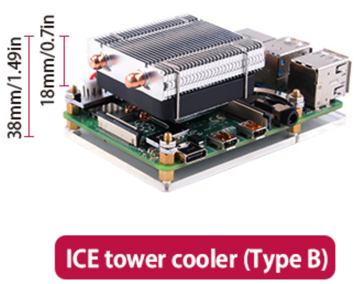

- Compare with other products

Compare with other products from Dimention

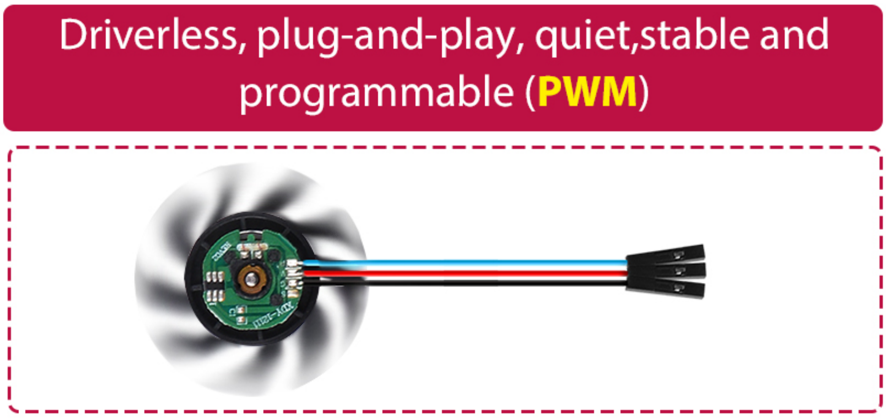


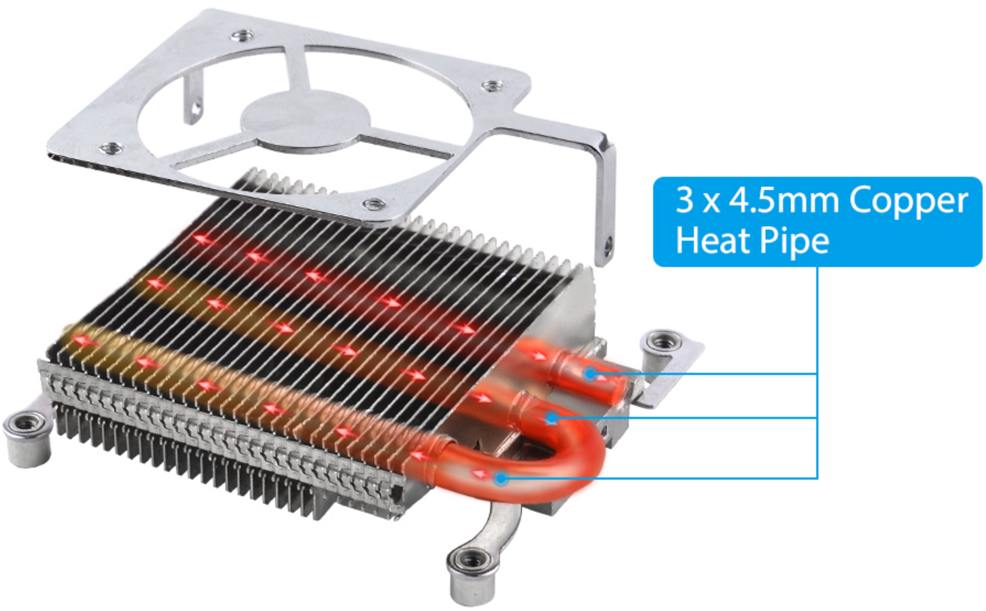




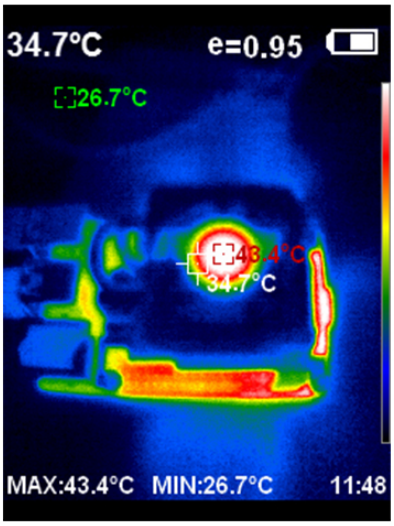



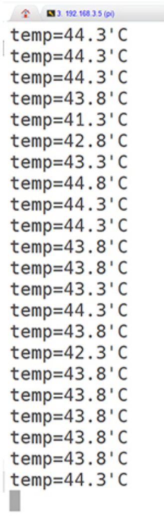

- Structure of the heat dissipation system




- Benchmark by using sysbench on Raspberry Pi 64Bit OS.



```
temp=44.3'C
temp=44.3'C
temp=44.3'C
temp=43.8'C
temp=41.3'C
temp=42.8'C
temp=43.3'C
temp=44.8'C
temp=44.3'C
temp=44.3'C
temp=43.8'C
temp=43.8'C
temp=43.3'C
temp=44.3'C
temp=43.8'C
temp=42.3'C
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temp=43.8'C
temp=43.8'C
temp=43.8'C
temp=44.3'C
```
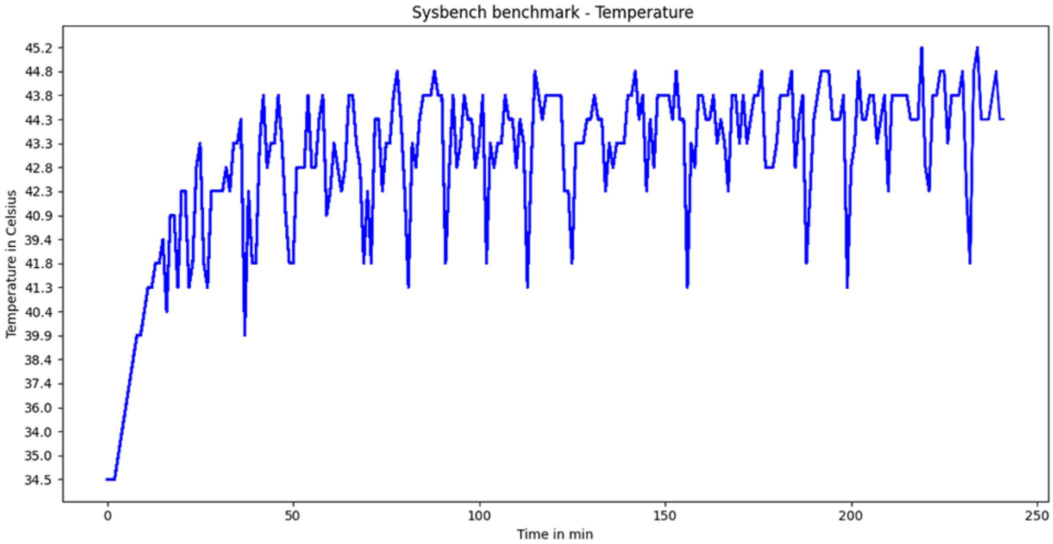
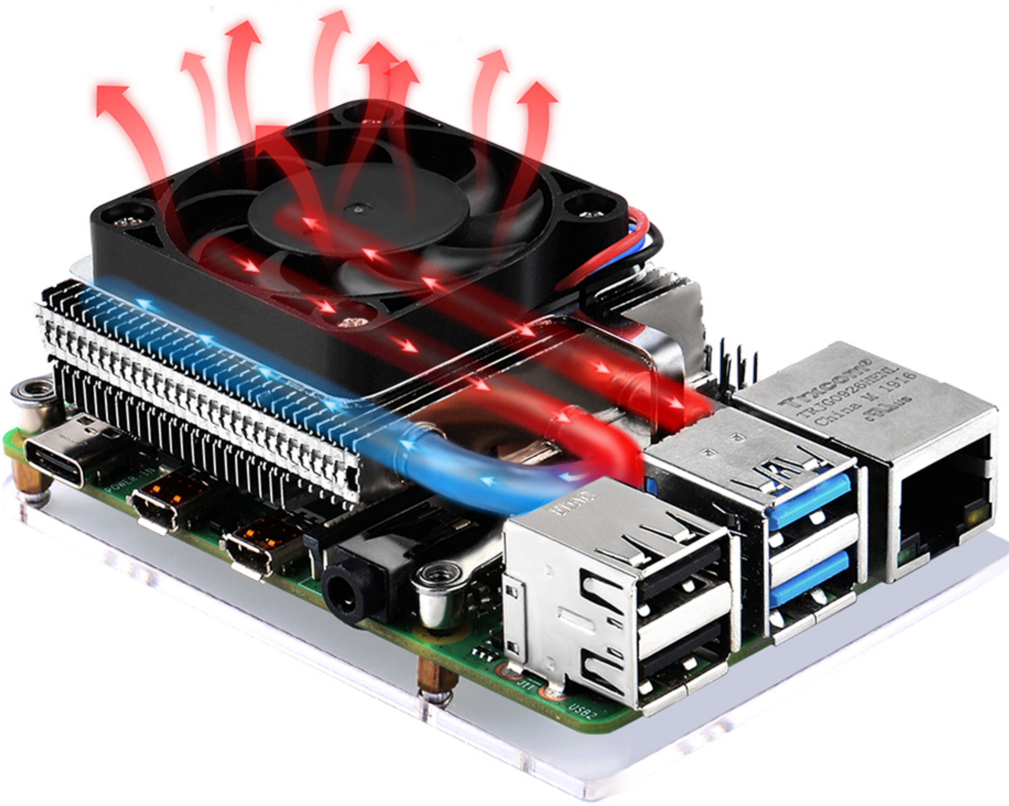
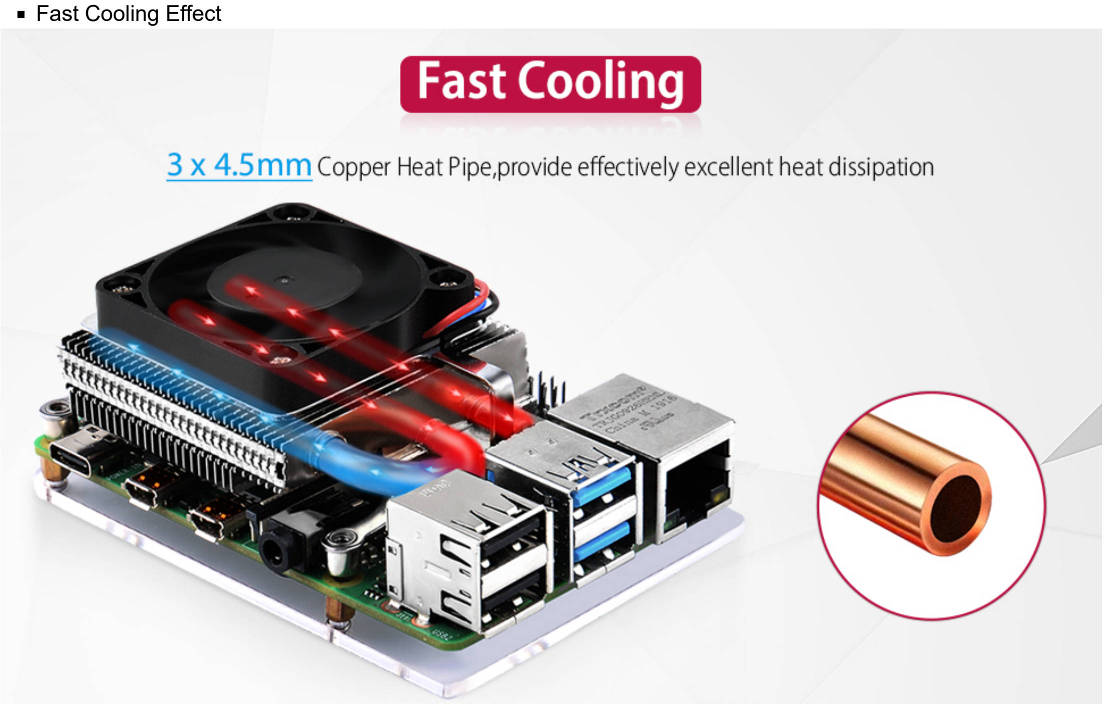





- Heat Dissipation

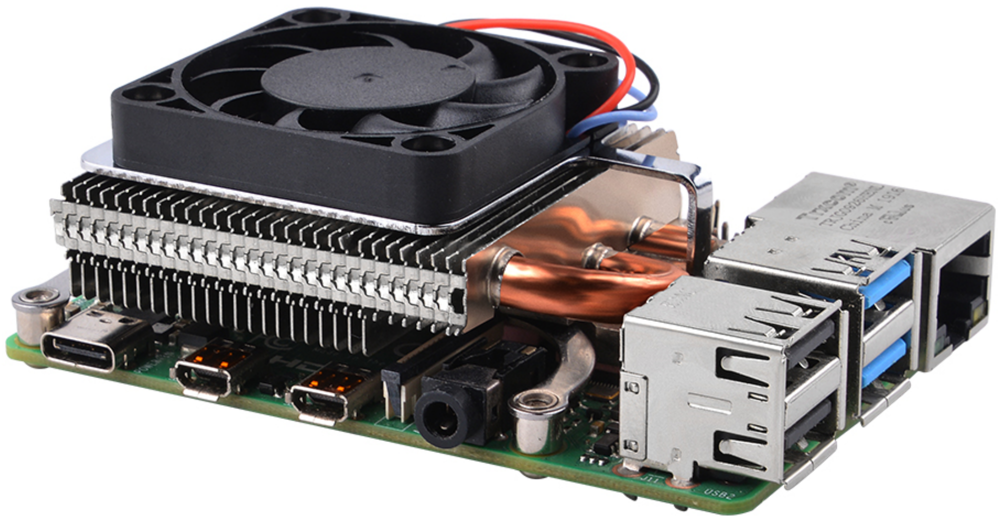


- Fast Cooling Effect

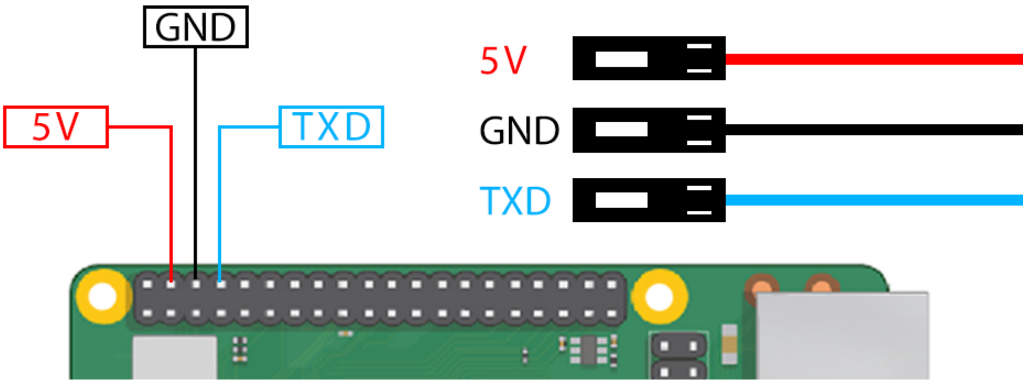



- Details

- Other side

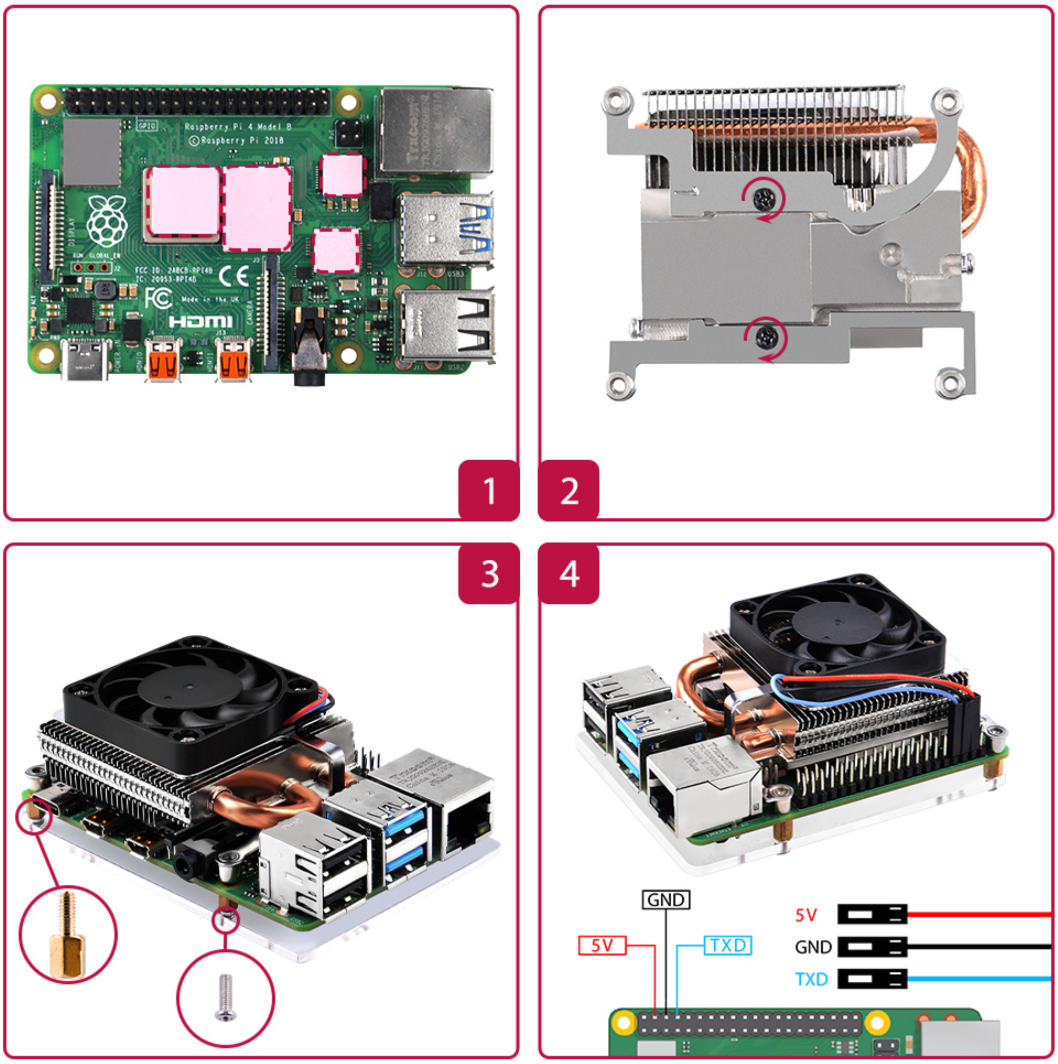


- Wiring



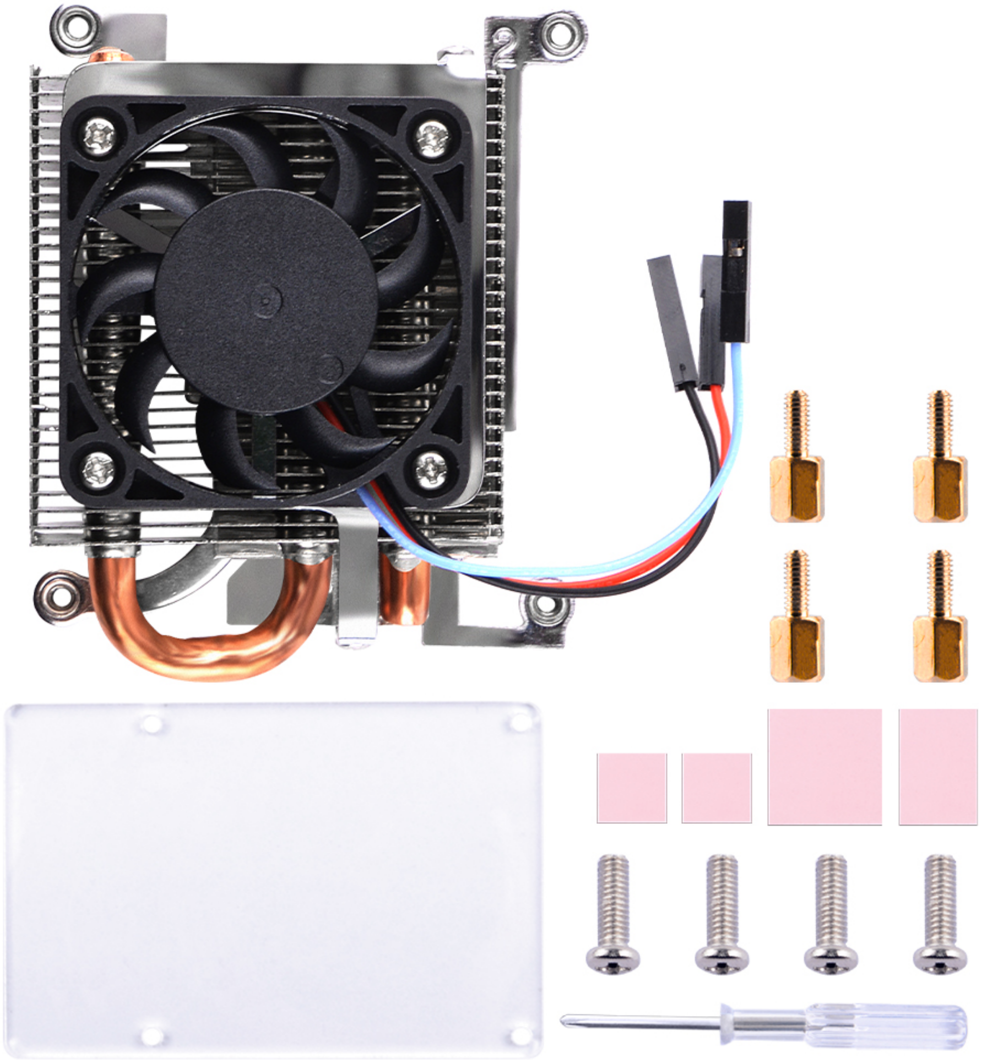

## How to assemble

- 1. Fix the bracket to ICE tower cooler with M2.5 screws.
- 2. Peel off the protect film of thermal pad and paste it on top of CPU and Memory chip.
- 3. Fix the copper pillar to the bracket.
- 4. Fix ICE tower cooler to Raspberry Pi 4B with M2.5 nuts.
- 5. Connect Red wire to Raspberry Pi's GPIO on 5V
- 6. Connect Black wire to Raspberry Pi's GPIO on GND
- 7. Connect Blue wire to Raspberry Pi's GPIO on TXD




## Package Includes

- 1 x Ultra Thin ICE Tower Cooler
- 4 x M2.5 Screws
- 4 x Copper Pillar
- 1 x Screw driver
- 4 x Thermal Pad
- 1 x Acrylic panel



## How to control fan speed

- Default OS: Raspberry Pi OS
- GPIO 14 is connected to fan pwm controller pin, you can just testing by using this code and modify it as you well.
- Create a file named fan_control.py
- Copy and paste following code

```
import RPi.GPIO as GPIO
import time
import subprocess as sp


# initializing GPIO, setting mode to BOARD.
# Default pin of fan is physical pin 8, GPIO14
Fan = 8
GPIO.setmode(GPIO.BOARD)
GPIO.setup(Fan, GPIO.OUT)

p = GPIO.PWM(Fan, 50)
p.start(0)

try:
    while True:
        temp = sp.getoutput("vcgencmd measure_temp|egrep -o '[0-9]*\.[0-9]*'")
        # print(temp)
        if float(temp) < 48.0:
            p.ChangeDutyCycle(0)
        elif float(temp) > 48.0 and float(temp) < 60.0:
            p.ChangeDutyCycle(100)
            time.sleep(0.1)
        elif float(temp) > 60.0:
            p.ChangeDutyCycle(100)
            time.sleep(0.1)

except KeyboardInterrupt:
    pass
p.stop()
GPIO.cleanup()
```

- Save it and test it

```
python3  fan_control.py
```

## Keywords

- ICE tower cooler, ultra thin ice tower cooler, ice tower for raspberry pi 4B